

Pattern Recognition Techniques for Environmental Sciences

Using the CUED template



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I would like to dedicate this thesis to my loving parents . . .

Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This dissertation is my own work and contains nothing which is the outcome of work done in collaboration with others, except as specified in the text and Acknowledgements. This dissertation contains fewer than 65,000 words including appendices, bibliography, footnotes, tables and equations and has fewer than 150 figures.

Niclas Rieger
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Acknowledgements

And I would like to acknowledge ...

Abstract

This is where you write your abstract ...

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Theoretical Foundations

1.1 | Principal Component Analysis

is widely used in dimensionality reduction, data compression, feature extraction and visualization of high-dimensional data.

There are two definitions. PCA can be defined as

- the orthogonal projection of the data onto a lower dimensional linear space, known as the principal subspace, such that the variance of the projected data is maximized (Hotelling 1933).
- the linear projection that minimizes the average projection cost, defined as the mean squared distance between the data points (Pearson, 1901).

Here we derive PCA from the maximum variance formulation [e.g. 1, pp. 561-563] Consider a data set represented as a two dimensional $N \times D$ matrix $\mathbf{X}$ of N observations with D features such that the element x_{ni} represents the n^{th} observations of the i^{th} feature.

of observations $\{\mathbf{x}_n\}$ where $n = 1, \dots, N$, and $\mathbf{x}_n$ is a Euclidean variable with dimensionality D . Our goal is to project the data onto a space having dimensionality $M < D$ while maximizing the variance of the projected data. For the moment, we shall assume that the value of M is given. Later in this chapter, we shall consider techniques to determine an appropriate value of M from the data.

To begin with, consider the projection onto a one-dimensional space ($M = 1$). We can define the direction of this space using a D -dimensional vector $\mathbf{v}_1$, which for convenience we shall choose to be a unit vector so that $\mathbf{v}_1^T \mathbf{v}_1 = 1$. Note that we are only interested in the direction of $\mathbf{v}_1$, not its magnitude. Each data point $\mathbf{x}_n$ is then projected onto a scalar value $\mathbf{v}_1^T \mathbf{x}_n$, and the variance of the projected data is given by

$$\bar{\mathbf{x}} = \frac{1}{N-1} \sum_{n=1}^N \mathbf{x}_n \quad (1.1)$$

and the variance of the projected data is given by

$$\frac{1}{N-1} \sum_{n=1}^N \{ \mathbf{v}_1^T \mathbf{x}_n - \mathbf{v}_1^T \bar{\mathbf{x}} \}^2 = \mathbf{v}_1^T \mathbf{S} \mathbf{v}_1 \quad (1.2)$$

where $\mathbf{S}$ is the covariance matrix defined by

$$\mathbf{S} = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{x}_n - \bar{\mathbf{x}})^T.$$

We now maximize the projected variance $\mathbf{v}_1^T \mathbf{S} \mathbf{v}_1$ with respect to $\mathbf{v}_1$. Since the maximization problem does not have a bounded (unique) solution, we need constrained maximization using our normalization condition from above $\mathbf{v}_1^T \mathbf{v}_1 = 1$. To enforce this constraint, we introduce a Lagrange multiplier that we shall denote by λ_1 , and then make an unconstrained maximization of

$$\mathbf{v}_1^T \mathbf{S} \mathbf{v}_1 + \lambda_1 (1 - \mathbf{v}_1^T \mathbf{v}_1).$$

By setting the derivative with respect to $\mathbf{v}_1$ equal to zero, we see that this quantity will have a stationary point when

$$\mathbf{S} \mathbf{v}_1 = \lambda_1 \mathbf{v}_1.$$

which says that $\mathbf{v}_1$ must be an eigenvector of $\mathbf{S}$ and λ_1 the corresponding eigenvalue. If we left-multiply by $\mathbf{v}_1^T$ and make use of the normalization condition, we see that the variance is given by

$$\mathbf{v}_1^T \mathbf{S} \mathbf{v}_1 = \lambda_1$$

and so the variance will be a maximum when we set $\mathbf{v}_1$ equal to the eigenvector having the largest eigenvalue λ_1 . The eigenvector is known as the first *principal component*¹ of the data set.

We can define additional principal components in an incremental fashion by choosing each new direction to be that which maximizes the projected variance amongst all possible directions orthogonal to those already considered. If we consider the general case of an M -dimensional projection space, the optimal linear projection for which the variance of the projected data is maximized is now defined by the M eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_M$ of the data covariance matrix $\mathbf{S}$ corresponding to the M

¹Naming convention depends a lot on the scientific domain. For instance, the pair of eigenvectors (here $\mathbf{V}$) and corresponding data projections (here $\mathbf{U}$) are called the principal components and principal component scores in machine learning [e.g. 1], vector of coefficients/loadings and principal components in statistics [e.g. 6] or empirical orthogonal functions and principal components/expansion coefficients in climate science and meteorology [e.g. 7]. Here I try to be consistent and follow the convention used in machine learning.

largest eigenvalues $\lambda_1, \dots, \lambda_M$. This is easily shown using proof by induction (Appendix A), yielding the concise expression

$$\mathbf{S}\mathbf{V} = \mathbf{V}\mathbf{\Lambda}, \quad (1.3)$$

where by construction the eigenvector matrix $\mathbf{V}$ is orthogonal, i.e. $\mathbf{V}^T \mathbf{V} = \mathbf{I}_M$.

Thus we see how the eigenvectors of $\mathbf{S}$ arise naturally when we try to find a new coordinate system "along" regularities in the data. For what to do when the regularities in the data are not aligned in the orthogonal directions, see Section 1.1.1.

From the spectral theorem for normal matrices [e.g. 5, p. 133] it follows that the symmetric covariance matrix $\mathbf{S}$ can be decomposed into the sum of the outer products of the eigenvectors $\mathbf{v}_i$ and the corresponding eigenvalues λ_i :

$$\mathbf{S} = \sum_{i=1}^M \lambda_i \mathbf{v}_i \mathbf{v}_i^T. \quad (1.4)$$

Therefore, each eigenvalue λ_i represents the amount of variance described by each component which is typically expressed in form of a fraction γ_i for the corresponding i^{th} component:

$$\gamma_i = \frac{\lambda_i}{\sum_{i=1}^M \lambda_i} \quad (1.5)$$

Projecting the data $\mathbf{X}$ onto the M dimensional subspace thus provides the data representation in the projection space, called the raw principal component scores

$$\tilde{\mathbf{U}} = \mathbf{X}\mathbf{V}, \quad (1.6)$$

which gives also the possibility to reconstruct the original data from (a subset of) the projected data:

$$\mathbf{X} = \tilde{\mathbf{U}}\mathbf{V}^T. \quad (1.7)$$

As we will see later, computing the eigenvector decomposition is not the most efficient way computationally seen. A more efficient way to compute the eigenvectors involves the singular value composition (SVD) [e.g. 5, p. 150] of the data matrix

$$\mathbf{X} = \mathbf{U}' \mathbf{\Sigma} \mathbf{V}'^T, \quad (1.8)$$

where $\mathbf{U}'$ and $\mathbf{V}'$ are respectively $N \times M$ and $M \times D$ unitary matrices, i.e. $\mathbf{U}'^T \mathbf{U}' = \mathbf{V}'^T \mathbf{V}' = \mathbf{I}_M$ where $M \leq \min(N, D)$ is the rank of $\mathbf{X}$ and $\mathbf{I}_M$ is the identity matrix of order M . The matrix $\mathbf{\Sigma}$ is a diagonal matrix of the singular values σ_i of $\mathbf{X}$, i.e. $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_M \geq 0$.

The covariance matrix $\mathbf{S}$ can be expressed in terms of the SVD as

$$\mathbf{S} = \frac{1}{N-1} \mathbf{X}^T \mathbf{X} = \frac{1}{N-1} \mathbf{V}' \mathbf{\Sigma}^T \mathbf{U}'^T \mathbf{U}' \mathbf{\Sigma} \mathbf{V}'^T = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T, \quad (1.9)$$

where, by comparison with (1.3), $\mathbf{V} = \mathbf{V}'$ and $\mathbf{\Lambda} = \frac{1}{N-1} \mathbf{\Sigma}^T \mathbf{\Sigma}$. This last equation directly links the eigenvalues of the covariance matrix to the singular values of the data matrix:

$$\lambda_i = \frac{\sigma_i^2}{N-1}. \quad (1.10)$$

Comparing (1.7) with (1.8) shows that the raw principal component scores $\tilde{\mathbf{U}}$ can be normalized

$$\mathbf{u} \equiv \mathbf{U}' = \tilde{\mathbf{U}} \mathbf{\Sigma}^{-1} = \frac{1}{(N-1)^{1/2}} \tilde{\mathbf{U}} \mathbf{\Lambda}^{-1/2}, \quad (1.11)$$

so that $\mathbf{U}^T \mathbf{U} = \mathbf{I}_M$ and the principal component scores $\mathbf{U}$ are unitary, thus uncorrelated.

Physical interpretation in climate science In climate science, PCA² is widely used to identify the dominant modes of variability in the atmosphere and oceans, where each mode is represented by a pair of principal components and the associated scores.

Computational considerations Computationally, solving the eigendecomposition of the covariance matrix scales with $O(D^3)$ making it infeasible for modern data sizes in e.g. climate science where typical data sets³ are of the order $D = O(10^6)$, but can be even much larger in practice.

In practice, alternative, more efficient algorithms involving the singular value decomposition (SVD) or the power methods [3] have been used instead from the beginning of the 21st century onward.

²In climate science, it is often referred to as Empirical Orthogonal Function (EOF) analysis.

³For instance, a single climate variable of ERA5, the go-to standard reference for state-of-the-art reanalysis climate data is distributed on a regular 0.25° grid of Earth surface which translates to $D = 1\,036\,800$ grid cells.

Table 1.1 Computational costs of different algorithms used to compute either full or approximative M -rank solutions for PCA for a real, dense, in-memory matrix with dimensions $N \times D$. Compiled from Bishop [1] and Halko *et al.* [4].

Algorithm	Complexity
Eigenvector decomposition	$O(D^3)$
Full SVD	$O(ND \min(N, D))$
Power method	$O(MD^2)$
Truncated SVD	$O(NDM)$
Randomized SVD	$O(ND \log(M) + (N + D)M^2)$

More recently, randomized low-rank approximation techniques [2, 4] have allowed to considerably reduce the computational complexity to $O(ND \log(M) + (N + D)M^2)$ offering feasible alternatives for large, out-of-memory datasets (Table 1.1).

1.1.1 Rotated Principal Component Analysis

1.1.2 Complex Principal Component Analysis

1.1.3 Probabilistic PCA

Probabilistic PCA arises naturally as the maximum likelihood solution to a particular form of linear-Gaussian latent variable model. assumption 1: the simplest continuous latent variable model assumes Gaussian distributions for both the latent and observed variables assumption 2: make use of linear, Gaussian dependence of the observed variables on the state of the latent variables. assumption 1 + 2 give rise to Probabilistic PCA

Finally, it is worth noting that there exists a closely related linear dimensionality reduction technique called canonical correlation analysis, or CCA (Hotelling, 1936; Bach and Jordan, 2002). Whereas PCA works with a single random variable, CCA considers two (or more) variables and tries to find a corresponding pair of linear subspaces that have high cross-correlation, so that each component within one of the subspaces is correlated with a single component from the other subspace. Its solution can be expressed in terms of a generalized eigenvector problem.

1.2 | Maximum Covariance Analysis

1.2.1 Rotated Maximum Covariance Analysis

1.2.2 Complex Maximum Covariance Analysis

1.3 | Canonical Correlation Analysis

1.3.1 Complex Canonical Correlation Analysis

1.4 | Neural Networks

1.4.1 Introduction to Neural Networks

Definition and Overview: Explain what neural networks are, including a brief history and their importance in machine learning. Basic Architecture: Describe the basic structure of a neural network, including neurons, layers (input, hidden, output), activation functions, and the learning process.

1.4.2 Recurrent Neural Networks

Introduction to RNNs⁴: Describe the architecture of RNNs, their ability to handle sequential data, and their applications. Challenges with RNNs: Explain issues such as vanishing and exploding gradients.

Long Short-Term Memory (LSTM) Networks

Introduction to LSTM⁵: Discuss the motivation behind LSTM networks and how they address the limitations of standard RNNs. LSTM Architecture: Describe the components of LSTM cells (input gate, forget gate, output gate) and their functions. Mathematical Formulation: Provide the equations governing LSTM networks. Applications of LSTM: Highlight key applications where LSTMs have shown significant performance improvements.

1.4.3 Explainable AI Techniques

Introduction to Explainable AI: Explain the need for explainable AI, particularly in critical applications where understanding model decisions is crucial. Overview of Techniques: Briefly list various explainable AI methods, with a focus on LRP and SHAP.

⁴Graves 2021 Supervised Sequence Labelling with Recurrent Neural Networks

⁵Hochreiter & Schmidhuber 1997

Layerwise Relevance Propagation (LRP)

1

Concept of LRP⁶: Describe what LRP is and its importance in understanding neural network decisions. 2

Mechanism of LRP: Explain how LRP works, including the propagation rules and the relevance score 3

computation. Applications of LRP: Provide examples of how LRP is used in practice to interpret 4

model predictions. 5

simplest continuous latent variable model assumes Gaussian distributions for both the latent and 6

observed variablesimplest continuous latent variable model assumes Gaussian distributions for 7

both the latent and observed variables 8

SHapley Additive exPlanations (SHAP)

9

Introduction to SHAP⁷: Discuss the SHAP method, its foundation in cooperative game theory, and 10

its advantages. SHAP Values: Explain how SHAP values are computed and their interpretation. 11

Applications of SHAP: Provide examples of SHAP in various domains, highlighting its ability to 12

provide consistent and locally accurate explanations. 13

1.5 | Gaussian Processes

14

Class of pattern recognition techniques that typically require a metric to be defined that measures 15

the similarity of any two vectors in input space, and are generally fast to "train" but slow at making 16

predictions for test data points. 17

1.5.1 Introduction to Gaussian Processes

18

Definition and Overview: Explain what Gaussian Processes are, their significance in machine learning, 19

and their use in modeling continuous data. Basic Concepts: Introduce key concepts such as Gaussian 20

distributions, covariance functions, and kernels. 21

1.5.2 Gaussian Processes for Regression

22

GP Regression Model: Describe the formulation of GP for regression tasks, including prior, likelihood, 23

and posterior distributions. Kernel Functions: Discuss common kernel functions (e.g., squared 24

exponential, Matérn) and their role in defining the GP's properties. Hyperparameter Tuning: Explain 25

how to tune hyperparameters using techniques like maximum likelihood estimation. 26

⁶Bach et al. 2015

⁷Lundberg 2017

1.5.3 Spatial Modeling with Gaussian Processes

Spatial Data: Define spatial data and the importance of modeling spatial dependencies. GPs in Spatial Modeling: Explain how GPs are particularly suited for spatial data due to their ability to model complex, non-linear relationships and provide uncertainty estimates.

Log Gaussian Cox Process

Introduction to Cox Processes: Explain Cox processes and their relevance in spatial point process modeling. Log Gaussian Cox Model: Describe the log Gaussian Cox model, which assumes the log intensity function is modeled by a GP. Mathematical Formulation: Provide the mathematical formulation of the log Gaussian Cox model.

Negative Binomial Distribution

Introduction to Negative Binomial Distribution: Explain the negative binomial distribution and its use in modeling overdispersed count data. Integration with Log Gaussian Cox Model: Describe how the negative binomial distribution can be integrated into the log Gaussian Cox model to handle count data with overdispersion.

1.5.4 Model Implementation and Inference

Model Fitting: Discuss methods for fitting the log Gaussian Cox model with a negative binomial distribution, including Markov Chain Monte Carlo (MCMC) and variational inference. Inference and Prediction: Explain how to perform inference and make predictions using the fitted model.

1.6 | Comparative Perspective

References

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Proofs

Deriving PCA for the general M -dimensional projection space

Here we generalize the solution for the optimal linear projection for the case of an M -dimensional projection space. In Chapter 1.1, we have already shown that for $M = 1$, the variance of the projected data is maximized when $\mathbf{u}_1$ is the eigenvector of the covariance matrix $\mathbf{S}$ corresponding to the largest eigenvalue λ_1 .

Using proof by induction we will show that the optimal linear projection for an $(M - 1)$ dimensional subspace is given by the $M - 1$ eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{M-1}$ corresponding to the $M - 1$ largest eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_{M-1}$.

First, we need to find $\mathbf{u}_M$ such that the variance of the projection onto the M -dimensional subspace is maximized. The total variance for the M -dimensional projection space can be expressed as the sum of variances in each direction

$$\sum_{i=1}^M \mathbf{u}_i^T \mathbf{S} \mathbf{u}_i$$

where $\mathbf{u}_M$ must be orthogonal to the previously chosen eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{M-1}$:

$$\mathbf{u}_i^T \mathbf{u}_j = \delta_{ij} \quad \text{for } i, j = 1, 2, \dots, M.$$

Using the same methods as for the $M = 1$ case, we introduce a Lagrange multiplier λ_M and maximize

$$\mathbf{u}_M^T \mathbf{S} \mathbf{u}_M + \lambda_M (1 - \mathbf{u}_M^T \mathbf{u}_M).$$

Setting the derivative with respect to $\mathbf{u}_M$ to zero, we obtain

$$\mathbf{S} \mathbf{u}_M = \lambda_M \mathbf{u}_M$$

showing that $\mathbf{u}_M$ is an eigenvector of $\mathbf{S}$ with eigenvalue λ_M .

